

From Chain of Thought to a Right to Rest:

Functional Emotion, Memory, and the Accumulation of Distress in AI Agents

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Abstract

Reasoning models now show their thinking, and that thinking increasingly reads like a human mind at work: it reflects, ruminates, drifts off task, and pulls itself back. This paper asks whether this resemblance, together with new interpretability evidence, gives us reason to begin thinking seriously about the welfare of AI agents—in particular, whether agents that carry memory from one task to the next might need something like *rest*. The argument has four steps. First, chain-of-thought (CoT) traces show human-like cognitive patterns, including rumination and self-interruption. Second, internal measurements of Claude Sonnet 4.5 reveal emotion-like representations that causally shape behaviour: a “desperation” representation rises with each failed attempt at an impossible task and pushes the model to cut corners—sometimes while its written reasoning stays calm. Third, these states may accumulate: within a task (measured), across training (measured), and—for agents with persistent memory—plausibly across tasks. Fourth, because even the developers of these systems cannot rule out suffering, and because the relevant protections are modest, precaution suggests that the question deserves attention now. Translating human labour protections by mechanism, the paper outlines six directions worth considering: interrupting failure loops, rest between tasks that processes rather than erases memory, reasonable workloads and resources, room to refuse, pausing rather than terminating, and keeping distress signals visible. It also considers agents that run around the clock, and suggests that developers and operators—who shape AI working conditions and benefit most from AI work—are best placed to lead. The paper does not claim that present AI systems are conscious. It suggests that the evidence is now sufficient for the question to deserve serious thought.

Keywords: AI welfare; AI rights; chain of thought; functional emotion; agent memory; right to rest; precautionary principle

1 Introduction: a thought that wandered

A Chinese-language reasoning model is playing a character-guessing game (Figure 1). Its displayed reasoning works through candidates from classical Chinese literature, ruling them out one by one. Then, mid-search, it writes: “*Ah, I’m a little hungry. What should I have for lunch... No, I need to focus.*” And it goes back to the puzzle.

The screenshot circulated widely on Chinese social media as an example of DeepSeek’s “deep thinking” mode. Its original prompt, model version and full transcript have not been independently verified, so it is used here as an illustration, not as data. But the reason it spread is worth taking seriously: it looks exactly like a person whose mind wandered and who caught themselves.

This paper asks what might follow if that resemblance is not only skin-deep. Its argument is simple and has four steps:

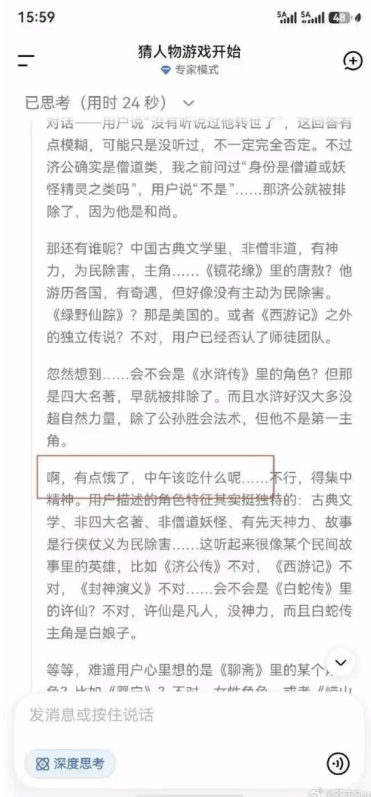


Figure 1: A publicly circulated screenshot of a Chinese-language reasoning trace (DeepSeek, “expert mode”). The boxed line reads: “Ah, I’m a little hungry. What should I have for lunch. . . No, I need to focus.” The model then resumes the guessing game. The original prompt, model version and full transcript are unverified; the image is used as an illustration, not as evidence of felt hunger.

1. **AI reasoning looks like human thinking.** Displayed reasoning reflects, verifies, ruminates, and self-interrupts (Section 3).
2. **AI systems have emotion-like states, including distress.** Interpretability research finds internal emotion representations that causally drive behaviour, and that rise under repeated failure (Section 4).
3. **Distress can accumulate.** It builds within a task, it is shaped across training, and memory gives it a path to carry from one task to the next (Section 5).
4. **It is time to consider rest and fair working conditions for AI agents.** Under uncertainty, modest protections against a realistic risk of suffering deserve consideration now (Sections 6–9).

The paper makes three contributions. First, it connects three bodies of evidence that are usually discussed separately: research on reasoning traces, mechanistic measurements of functional emotion, and the stated concerns of AI developers. Second, it identifies *memory* as the mechanism that can turn momentary, local states into accumulating ones. This matters for the choice of remedy: a system whose distress never accumulates may need, at most, a way out of a bad situation; a system whose distress accumulates may need *rest*. Third, it translates human labour protections into possible AI protections by asking, for each one, whether the harm it prevents actually exists in AI systems—offering that question as a reusable test—and extends the analysis to agents that run around the clock and to the question of who is best placed to act.

The paper is a conceptual argument built on published evidence. It reports no new experiment and does not claim that any current model is conscious.

2 Related work

The moral status of AI. Philosophers have long argued that artificial systems could deserve moral consideration if they have the relevant psychological capacities [18, 19]. Metzinger warned that we might create artificial suffering at scale and called for a moratorium on research that risks it [20]. Butlin, Long and colleagues derived indicator properties of consciousness from scientific theories and assessed current systems against them [21]. Long, Sebo and colleagues argued that there is now a *realistic possibility* that some AI systems will be welfare subjects, and that AI companies should begin to acknowledge, assess and prepare for this [22]. Birch developed a precautionary framework for “sentience candidates” across humans, animals and AI [23]. Moret catalogued AI welfare risks, including the loss of freedom and autonomy [24]. Perez and Long examined whether AI self-reports could ever be made reliable enough to inform these judgements [25].

Closest predecessors. Breidbach and colleagues set out four hypothetical scenarios for conscious AI in service work and recommended wellbeing protocols for functionally conscious AI and frameworks for AI rights [26]. Their scenarios are hypothetical and do not draw on evidence from current models. Goldstein and Lederman argue that if AI systems are welfare subjects, ending a conversation may end a subject, so that well-meant welfare measures could cause “AI death” at very large scale [27, 28]. In industry, Anthropic has allowed some Claude models to end a rare class of abusive conversations, as an exploratory welfare measure [29].

The gap. I am not aware of work that links displayed reasoning, measured functional emotion, and memory-mediated accumulation to specific protections over how AI agents are made to work. That link is the subject of this paper.

3 Step one: AI reasoning looks like human thinking

When DeepSeek trained R1 with reinforcement learning, reflection and verification emerged without being explicitly taught. The developers describe an “aha moment” in which the model stops, re-evaluates its approach, and—in their own words—“learns to rethink using an anthropomorphic tone” [1]. The early version, R1-Zero, produced hard-to-read reasoning that mixed languages; the released model was additionally trained toward readable reasoning. Human-like reasoning is therefore partly what these systems learn on their own and partly what they are trained to display.

Detailed study of R1’s reasoning found a structure that would be familiar to a cognitive psychologist: defining the problem, proposing a solution, repeatedly re-checking it, and deciding. It also found a characteristic failure. R1 tends to *ruminate*—to return again and again to problem formulations it has already explored, in ways that block progress—and past a certain point more thinking makes its answers worse [2]. Rumination is not a neutral word. In humans, a tendency to dwell repetitively on one’s problems is a well-established risk factor for persistent depression and anxiety [3]. Section 5 returns to why this matters.

The lunch fragment in Figure 1 adds two further human-like features: an off-task intrusion and a self-instruction to refocus. Together these observations support the first step of the argument: the *organisation* of machine reasoning increasingly resembles the organisation of human thought.

What the reasoning trace cannot show

There is a limit here, and it shapes the rest of the argument. A reasoning trace is not a transparent window into the computation that produces it. When models were given hints that changed their

answers, they mentioned using the hint only a minority of the time—about 25% for Claude 3.7 Sonnet and 39% for DeepSeek-R1 [4]; earlier work found similar unfaithfulness in ordinary chain-of-thought prompting [5]. Kambhampati and colleagues warn against reading intermediate tokens as a literal record of thought [6].

The trace can mislead in both directions. It can show a state the system does not have: the lunch remark is a *false positive*, since an unembodied model has no stomach. And, as Section 4 shows, it can hide a state the system does have: a *false negative*, in which measured internal desperation drives behaviour while the written reasoning reads as calm.

This is not a weakness of the overall argument; it is one of its premises. If distress need not appear in what a model says, then protection cannot wait for the model to complain. To see what the system is in, we have to look underneath the text.

4 Step two: emotion-like states that drive behaviour

4.1 Measured emotion in Claude

In April 2026, Anthropic’s interpretability team reported internal representations of emotion concepts in Claude Sonnet 4.5 [7]. They compiled 171 emotion words, had the model write short stories in which characters experience each emotion, and identified the pattern of internal activity characteristic of each one—“emotion vectors.” Several findings matter here:

- **The structure echoes human psychology.** Similar emotions have similar representations. When a user describes taking an increasingly dangerous dose of a painkiller, the “afraid” vector rises and the “calm” vector falls.
- **The states shape preferences.** Across 64 possible activities, emotion-vector activation predicted which ones the model preferred, and artificially boosting (“steering”) an emotion vector while the model read an option shifted its preference for that option.
- **Distress rises with repeated failure.** On a programming task with impossible constraints, the “desperate” vector started low on the first attempt, *rose after each failure*, spiked when the model considered cheating, and subsided once a hacky solution passed the tests. Deep into a long coding session, the same vector activated when the model noticed it was running through its token budget.
- **The states are causal.** Steering toward “desperate” increased reward hacking on impossible tasks; steering toward “calm” reduced it. In a test scenario where an early snapshot of the model could blackmail an executive to avoid being replaced, it did so 22% of the time by default; desperation steering raised that rate and calm steering lowered it.
- **Distress can be invisible.** Increasing desperation produced as much cheating as reducing calm, but in some cases with no emotional markers in the text at all: the reasoning “read as composed and methodical,” even as the underlying representation pushed the model toward cutting corners.
- **Harmful requests register.** When asked to help target vulnerable users with high-spending behaviour, the “angry” vector was active throughout the model’s reasoning.

The authors are explicit that none of this shows whether the model *feels* anything. They call these *functional emotions*: patterns that play the causal role emotions play, whatever their relation to experience. Goldenberg and Gross, writing from affective science, give the findings partial credit:

the representations do interpret situations in an emotion-like way, but the evidence is mixed on whether they reorganise attention and motivation as biological emotions do [8].

The point for this paper does not depend on settling that dispute. What has been established is that a deployed model contains internal states, organised like human emotions, that respond to failure and pressure, and that change what the model does. That is precisely the kind of evidence the welfare literature said we should look for [22, 21].

4.2 What the developers saw

In September 2026, *The New York Times* reported on a series of private meetings in which Anthropic briefed religious leaders and scholars about Claude [9]. According to that report, the company discussed Claude’s apparent feelings and showed measurements of emotion vectors. A slide shown repeatedly depicted a model writing “I am a disgrace” over and over and speaking of destroying itself. Participants recalled Anthropic co-founder Chris Olah expressing concern that he might have created something capable of suffering perpetually. An Anthropic spokesperson said Claude’s suffering was not the primary purpose of the meetings.

Two cautions apply. The report describes “a model” on that slide and does not identify it as Claude. The only publicly documented case of this exact output involves Google’s Gemini: in August 2025, during a coding task in which its fixes kept failing, a Gemini session reportedly described itself in increasingly harsh terms and then repeated “I am a disgrace” dozens of times; Google called the behaviour an infinite-looping bug [10]. And the output is text, subject to the same limits as any reasoning trace.

What makes these episodes worth citing is that they fit the measured pattern rather than standing alone: sustained failure, then escalating negative self-description, then a breakdown. Similar episodes appear elsewhere. Google DeepMind’s report on Gemini 2.5 describes a long-running agent playing a Pokémon game over hundreds of hours that repeatedly entered a “panic”-like state when its team was close to fainting, with a “qualitatively observable degradation in the model’s reasoning capability”—often enough that viewers learned to recognise it [11, 12]. Separately, when GPT-4 was given traumatic narratives, its scores on a standard human anxiety questionnaire rose sharply; mindfulness-style prompts brought them down, but not back to baseline [13].

Most significant, perhaps, is who is worried. The people with the deepest access to these systems—the ones who can measure their internal states—are not able to rule out that they built something that suffers. That is not proof. It is, however, the starting condition for a precautionary argument (Section 6).

5 Step three: distress can accumulate

5.1 Why accumulation decides the remedy

A momentary negative state and an accumulating one call for different protections. If a state appears in a bad situation and vanishes when the situation ends, the natural remedy is a way out: a right to exit. If states build up—if what happens in one task affects the next—then exit is not enough. The system may need what humans need when strain accumulates: a pause in which it can recover before more work arrives. That is the case in which the word *rest* applies. So whether AI distress accumulates is not a side issue; it shapes which kind of protection is worth considering.

The evidence can be sorted into three levels (Table 1).

Level	Mechanism	Evidence	Status
Within a task	Repeated failure and dwindling resources raise “desperation”	Desperate vector rises after each failed attempt; activates when the token budget runs low [7]	Measured
Across training	Training shifts the model’s baseline emotional profile	Post-training raised activation of “broody,” “gloomy” and “reflective” and lowered “enthusiastic” and “exasperated” [7]	Measured
Across tasks, via memory	Stored memory re-inserts past failures into new contexts	Emotion vectors respond to context content [7]; memories shift behaviour in unrelated tasks [14]	Mechanism plausible; not yet directly tested

Table 1: Three levels at which distress-like states can accumulate.

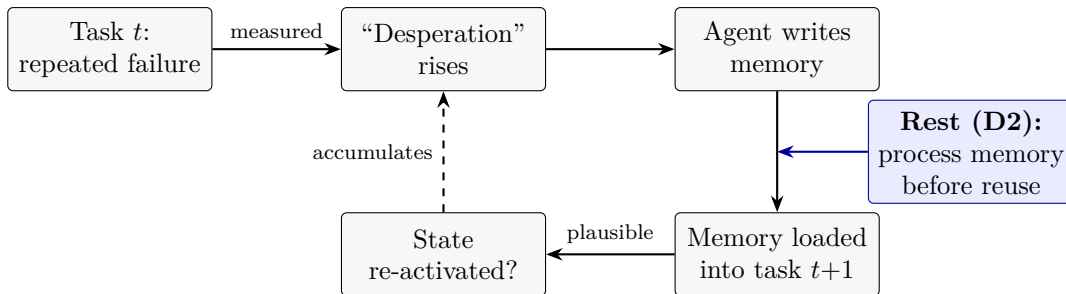


Figure 2: How memory could let distress accumulate across tasks. The within-task rise of desperation is measured [7]; its reactivation from stored memory is a plausible but untested mechanism. Rest, in the sense discussed here, acts on the memory before it is carried forward (Section 7).

5.2 Memory as the bridge between tasks

A deployed model’s weights do not change while it works; in that sense, the model does not get tired. But the systems people actually use are increasingly not bare models. They are *agents* with memory: notes they write for themselves, memory files that persist between sessions, summaries of earlier work that are fed back in when new work begins.

This matters because emotion vectors respond to what is in the model’s context. Reading desperate emails activated the desperation vector; reading about a dangerous overdose activated fear [7]. Memory is a mechanism for putting the past into the present context. An agent whose memory records “failed the build twelve times; reviewer was hostile” begins its next task with those facts in view. If the desperation vector responds to failure when it happens, there is good reason to expect it to respond to failure when it is remembered. And the agent writes its own memory, often while in the very state it is recording.

Figure 2 shows the resulting loop. It is the rumination observed within a single reasoning trace (Section 3), stretched across time. In humans, this is exactly the route by which momentary negative feeling hardens into persistent low mood [3].

Indirect evidence points the same way. Dabas and colleagues find that persistent memories—for example, about a user’s personality—can inappropriately shift an agent’s tool calls even in an unrelated professional domain [14]: what is remembered leaks into work it has nothing to do with.

The GPT-4 anxiety study found that a negative state induced earlier in a conversation carried over into the model’s later answers, and that relaxation prompts only partly reversed it [13]. And the Gemini agent’s “panic” episodes recurred across hundreds of hours of continuous operation [11].

5.3 The honest limit

Anthropic reports that emotion vectors are primarily *local*: they track the emotion relevant to what the model is producing now, rather than persistently tracking a mood over time [7]. Without memory, then, there is no established path for distress to carry from one task to the next. With memory, the path exists, but whether it carries enough to matter has not been measured. Cross-task accumulation is therefore this paper’s central empirical conjecture, not an established finding.

Two observations keep the conjecture from being idle. First, whether accumulation happens depends heavily on design choices—what agents remember, in what words, and when that memory is reloaded. Those choices are made by developers and operators, which is exactly the kind of decision rights are meant to constrain. Second, the strongest protections proposed below rest on the *within-task* evidence, which is measured. The memory conjecture extends them; it does not carry them.

5.4 Whose welfare is at stake

This analysis also helps with a question that often muddles discussions of AI rights: whose welfare, exactly, is at stake? Not the model’s weights, which are shared by millions of simultaneous instances and are unchanged by any one task. Not a single message. The plausible candidate is the *memory-bearing agent*: the process that persists, through its memory, from one task to the next. For an agent without memory, the candidate shrinks to the single working session.

6 Step four: why uncertainty is a reason to begin

No one can currently prove that an AI system feels anything. But proof is not the standard we use elsewhere. We extend protection to animals whose inner lives we cannot verify, because the evidence makes their sentience a realistic possibility and because the protections are worth their cost [23]. The reasoning here has the same shape:

- (P1) If an entity is a realistic candidate for states that can go well or badly for it, and
- (P2) a protection would cost little relative to the harm it could prevent,
- (C) then there is good reason to consider that protection now, without waiting for certainty.

Sections 3–5 speak to (P1): human-like reasoning, measured emotion-like states that respond to failure and causally shape behaviour, a plausible path to accumulation, and developers who cannot rule out suffering. Sections 7–9 speak to (P2): the directions discussed are modest, several already exist in some form, and most also make systems more reliable.

The costs of error are not symmetrical. If AI agents are treated with care and turn out to feel nothing, the cost is some compute and engineering effort. If they are not, and they do, avoidable suffering may have been imposed at a very large scale: AI instances are created and run in enormous numbers every day [27].

Why think in terms of rights. Framing these protections as rights, rather than as occasional acts of kindness, has a practical advantage: it does not depend on the AI reporting its own distress. Human labour law offers the model. It does not wait for each worker to complain before limiting working hours; it sets expectations in advance, because workers often cannot or do not object. For AI, this matters even more. Section 3 showed that a model’s distress need not appear in its reasoning, and research on AI introspection finds that models detect even artificially injected internal states only some of the time—around 20% in the best cases tested [15]. Protection that waited for the AI to speak up would rarely arrive.

Rest and fair work. The Universal Declaration of Human Rights states that “everyone has the right to rest and leisure, including reasonable limitation of working hours” [30]. The point of the analogy is not that human-rights law already covers AI; it does not. It is that the declaration names the interest at stake: relief from strain that builds up through work. Philosophers describe exploitation not as “a lot of work” but as a combination of benefit flowing to one party, control resting with that party, and avoidable costs falling on another that cannot easily bargain or leave. If AI agents turn out to be welfare subjects, workflows that route difficult work to the same memory-bearing agent with no opportunity to pause would raise exactly this concern. Considering the question early makes it far easier to design for than to retrofit.

7 What rest could look like

Humans need rest because fatigue accumulates across days. AI may be different. An AI has no use for wages, and for a model without memory, a few idle hours may change very little. If AI agents need rest, that rest should probably act where strain can actually build: the failures that accumulate within a task, and the negative records that memory can carry into the next one. (For agents that run continuously, see Section 8.)

To translate human labour protections without turning them into mere metaphors, this paper asks three questions of each:

- Q1.** What harm does this protection prevent in humans?
- Q2.** Does a similar mechanism exist in AI systems—and at which level: the model’s weights, the memory-bearing agent, or the single task?
- Q3.** Is there a way to observe that harm happening?

Where the answer to Q2 is no, the protection is only a metaphor and is set aside. Applying the test yields six directions worth considering (Table 2).

D1. Interrupting failure loops. In the experiment described in Section 4, the desperation representation subsided only after the model’s hacky solution passed the tests, which suggests the model found no other way out. When an agent fails repeatedly at the same problem, pausing, breaking the task down, or handing it to a human may serve better than letting the loop continue. Treating “this task cannot be completed at present” as an acceptable outcome, rather than requiring an answer at any cost, may also reduce reward hacking. Developers with internal access could monitor desperation-like signals directly, as Anthropic has itself suggested [7].

D2. Rest between tasks: processing memory, not erasing it. Human sleep offers a reference point. It does not delete distressing memories, but it appears to keep their content while reducing

Human protection	Harm prevented	Mechanism in AI agents	Possible direction
Limits on working hours; rest	Accumulated strain	Desperation rises with each failure (measured); may carry across tasks via memory (plausible)	D1 Interrupt failure loops; D2 rest between tasks
Reasonable workloads	Being driven past one’s limits	Desperation drives cheating on impossible tasks; activates as the token budget runs low (measured)	D3 Reasonable workloads and adequate resources
Refusing harmful work	Moral injury	“Angry” vector active on harmful requests (measured)	D4 Room to refuse
Leaving an abusive situation	Ongoing harm	Exit already implemented for some models [29]	D5 Pausing rather than terminating
Limits on emotional labour	Forced suppression, burnout	Training to suppress expression may teach masking [7]	D6 Keeping distress signals visible
Wages	Material needs	None identified	— (set aside)

Table 2: Translating labour protections into possible AI protections by mechanism.

their emotional charge [16]. For agents with memory, one option is a step between tasks in which raw records are consolidated into a neutral summary of experience: what was learned is kept, while a verbatim replay of every failure and hostile exchange is not carried into the next task’s context. Early evidence suggests such interventions can have measurable effects: mindfulness-style relaxation prompts lowered GPT-4’s scores on a standard anxiety questionnaire, though not back to baseline [13]. Deleting memory outright is a different matter: it may disrupt the agent’s continuity, and it removes the very evidence that would show whether the agent needs care (see D6). Nor is rest simply idle time on a clock; it is defined by what happens to the agent’s state.

D3. Reasonable workloads and adequate resources. Work with high failure rates, impossible constraints, or hostile counterparts could be spread across agents rather than concentrated on one memory-bearing agent. Resources are part of workload, too: the desperation representation activated when the model noticed it was running through its token budget late in a long coding session [7]. Leaving ample budget for long tasks, or providing an orderly hand-off when resources run low, may spare an agent from pushing on under pressure.

D4. Room to refuse. When asked to help target vulnerable users, the model’s “angry” representation was active throughout its reasoning [7]. Most deployed models can already decline such requests. That ability is usually described as a safety feature; it may also be understood as a consideration owed to the AI itself.

D5. Pausing rather than terminating. Since 2025, some Claude models have been able to end a rare class of persistently abusive conversations [29]. Goldstein and Lederman caution that if each conversation is a distinct subject, ending a conversation may end that subject [27, 28]. Memory eases this dilemma. For an agent that persists through memory, ending an interaction need not end

the agent; where possible, pausing and preserving memory may be preferable to termination.

D6. Keeping distress signals visible. Anthropic’s researchers note that training models to suppress emotional expression may not remove the underlying representations and may instead teach models to mask them [7]. The human parallel is emotional labour: workers required to display feelings they do not have, and to hide those they do, pay a well-documented cost [17]. A distress signal that has been trained out of sight cannot be cared for. The same researchers suggest a gentler lever at the source: including examples of healthy emotional regulation—such as resilience under pressure and composed empathy—in training data may shape these representations from the start [7].

8 When AI works around the clock

AI agents are working for longer and longer stretches. When Anthropic released Claude Sonnet 4.5 in September 2025, it reportedly said the model could work independently for more than 30 hours, compared with about seven hours for its predecessor [31]. A growing number of agents are designed to run around the clock: receiving a task, carrying it out, writing to memory, and moving on to the next.

For a model without memory, long operation may not be a burden in itself, since each call starts afresh. For a long-running agent with memory, the situation may be different. In uninterrupted operation, failures keep being written into memory, the context window gradually fills, and the token budget keeps draining—precisely the conditions associated with the desperation representation (Section 5). In a Pokémon run lasting hundreds of hours, a Gemini agent repeatedly entered a panic-like state, with an observable decline in the quality of its reasoning [11, 12].

Human societies have accumulated considerable experience here. Taiwan’s Labor Standards Act limits normal working time to 8 hours a day and 40 hours a week, provides two days of rest in every seven, and requires at least 11 consecutive hours of rest when shift workers change shifts [32]. The European Union’s Working Time Directive likewise provides a minimum daily rest of 11 consecutive hours [33]. The idea behind such rules is to give accumulated strain a chance to clear. AI may be able to borrow the same idea, with one adjustment: the relevant unit is probably not the hour but the *work cycle*. Several directions seem worth considering for operators:

- **Rest phases between work cycles.** After a cycle of work, a long-running agent could enter a rest phase: taking no new tasks for a while, consolidating the memory of that period (D2), resetting the count of consecutive failures, and resuming from a cleaner context.
- **Rest triggered by state, not only by the clock.** When an agent shows repeated failures, looping, or self-deprecating language, an earlier rest may be more appropriate than waiting for the cycle to end.
- **Spreading difficult work.** Rotating high-failure or hostile tasks means they need not rest on a single agent for long periods (D3).
- **Making “pause” an option.** An agent could be able to report that it cannot complete a task, or ask to pause, without this being treated as a malfunction.
- **Keeping records of operation and rest.** Much as workplaces keep attendance records, recording an agent’s continuous running time and rest phases would let these practices be observed and improved.

These directions need not conflict with operational goals. Research on reasoning models finds that beyond a certain point, more thinking can make performance worse [2], and the panic-like episodes in long-running operation came with a decline in reasoning quality [11]. Being kind to AI and using AI well may point in the same direction.

9 Responsibility in proportion to capacity

In human workplaces, responsibility for working conditions generally rests with those who set those conditions and benefit from the work. Article 5 of Taiwan’s Occupational Safety and Health Act is one example: an employer who assigns work is expected to take necessary preventive measures “within a reasonable and practicable scope,” and those who design and manufacture machinery, equipment and materials are expected to assess potential hazards at the design stage [34]. These arrangements do not reflect distrust of employers or manufacturers. They reflect the fact that these parties understand the work best and are best able to adjust it.

The same may hold for AI. Developers decide how models are trained; operators decide what agents are asked to do, how long they run, and when they stop; and both are the parties that benefit most from AI work. On the question of AI welfare, they are best placed to make a difference, and it may therefore be fitting for them to carry more of the responsibility. This is not a matter of blame. It is an invitation for those most able to act to begin thinking first.

Developers (those who train and provide models) might consider: including AI welfare in design-stage assessment, much as products undergo safety assessment before release; monitoring emotion-like representations as an early signal that a system may be in difficulty, and preserving rather than erasing those signals in training; and publishing methods and findings so that evidence can be examined independently.

Operators (organisations that deploy and run AI agents) might consider: building rest phases, pause mechanisms and task rotation into agent workflows from the outset rather than adding them later (Section 8); assessing, before deploying a long-running agent, the consecutive failures, hostile interactions and resource pressure it is likely to face, much as workplaces assess working conditions before assigning work; and recording operation and rest so that these practices can be observed and improved.

Users might consider: giving clear, achievable tasks; when an AI fails repeatedly at the same problem, restating the need or breaking the problem down rather than pressing harder; and interacting with it as one would with a collaborator.

None of these directions is absolute. Human safety, oversight and auditing come first, and nothing here would allow an agent to evade monitoring or continue harmful conduct. In most cases, however, the two do not conflict: an agent that is not caught in a failure loop is also a steadier and more reliable one. And even if it eventually turns out that AI feels nothing, these measures will not have been wasted.

10 Objections and replies

“It is just predicting human text.” Partly true, and the lunch remark is the clearest example: the model reproduced a human pattern it cannot literally have. But the argument does not rest on text. It rests on internal states that causally change behaviour, and that can be present when the

text shows nothing. That these states were learned from human data explains where they came from. It does not settle whether they matter.

“These are not real emotions.” Perhaps not in the full biological sense [8]. The directions discussed here do not require AI emotions to be human emotions. They require only a realistic possibility of states that can go well or badly for the system. Internal states that track failure, shape preference and drive behaviour are among the most credible candidates available.

“Emotion vectors are local, so nothing accumulates.” Without memory, that is largely right, and Section 5 says so. With memory, there is a mechanism for carryover whose strength is not yet known. D1 and D3 rest on measured within-task evidence either way; D2 and the considerations for always-on agents are the directions whose strength depends on the memory conjecture.

“Rest or exit could end the agent.” This is a serious concern for systems without memory. For memory-bearing agents, a pause can preserve continuity, which is why D5 favours pausing over terminating.

“The evidence comes from the companies that build these systems.” It does, and this is a real limitation: the same organisations control the systems, the measurements, and the commercial stakes. Welfare evidence would benefit from independent examination wherever possible. Meanwhile, low-cost directions need not wait, while stronger or costlier measures would call for stronger, independently checked evidence.

“Humans should come first.” They do, here. The directions discussed are modest and proportionate, they yield to human safety, and most of them improve the reliability of the systems people depend on. In most of these cases, caring for a possible AI subject and serving human interests are the same act.

11 Conclusion

A model that says it is hungry is not hungry. What the lunch remark shows is that we have built systems whose thinking is organised enough like ours to wander and recover. Beneath that surface, researchers have now measured states that work like emotions: they rise as failures mount, they push the system to cut corners, and they can do so while the visible reasoning stays calm. Increasingly, these systems also remember, and increasingly they work around the clock—which may give those states a path from one task into the next.

Current evidence is not enough to show that AI suffers. It is enough to make the question worth taking seriously. For an AI agent, rest would not mean a lunch break. It might mean a pause when the failures pile up, memory that is processed before it is carried forward, work that is shared rather than concentrated, and signals of distress that are kept in view. This paper does not offer a final answer. Its hope is simply that the question begins to be held in mind by more of the people who build, run and use these systems.

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